

Midterm Lab Exam React Native
Ui Navigation (Stack, Tab and Drawer)

General Instructions:

- Duration: 1hr and 30 minutes
- Tech Stack: React Native (Expo SDK54), TypeScript, Expo Router
- Project folder name: midterm_lab_exam_{surname}

Objective:

- Build a functional, multi-screen React Native application with simulated state authentication and nested navigation (Authentication Routing, Stack Navigation, Drawer Navigation, and Tab Navigation) using **Expo Router**.

Required App Directory Structure:

```
app/
├── (auth)/
│   └── index.tsx           # Welcome / Get Started Screen
├── (drawer)/
│   ├── (tabs)/
│   │   ├── _layout.tsx    # Tab Navigator Configuration
│   │   ├── details.tsx    # Tab Item: Details Screen
│   │   ├── index.tsx      # Tab Item: Main Dashboard / Home
│   │   ├── scan.tsx       # Tab Item: Scan Screen
│   │   └── settings.tsx   # Tab Item: Settings Screen
│   ├── _layout.tsx        # Drawer Navigator Configuration
│   ├── account-details.tsx # Drawer Item: Account Details Screen
│   ├── logout.tsx         # Drawer Item: Logout Trigger Screen
│   ├── profile.tsx        # Drawer Item: Profile Screen
│   └── _layout.tsx        # Root Layout (Handles Global Auth State)
```

Criteria for checking:

Part / Flow Section	Points	Specific Checklist (What the Prof will check)	Score
1. Authentication & Launch Flow	15 pts	Launch Screen Fidelity (5 pts): Displays the exact layout with the bold title, heart rate graphic icon, and active blue button.Get Started Action (5 pts): Clicking the button instantly moves the user forward into the core dashboard structure.History Stack Erasure (5 pts): Uses path replacing logic so a physical swipe back won't return the viewport to the launch screen.	
2. Bottom Tabs & FAB Interface	25 pts	News Feed Interface (5 pts): Correctly showcases the Facebook-style architecture with custom status components and interactive cards.Circular FAB QR Design (10 pts): Center tab is stylized as a bright blue, floating circle breaking upwards out of the normal tab boundary.QR Scan Tab Action (5 pts): Pressing the FAB circle cleanly displays the camera-ready dotted scan frame on the screen.Settings Core View (5 pts): The gear selector button on the right opens the structured system app preferences frame.	

3. Sidebar Drawer Menus	25 pts	Drawer Menu Header (5 pts): Sidebar renders the explicit custom top title text "Drawer Menus" smoothly.Home Redirection Link (5 pts): Selecting the top menu link safely restores focus to the user news feed home frame.My Profile Hook (5 pts): Clicking the second item triggers an exact route swap to pull up the individual details card page.Bottom Layout Logout (5 pts): The custom logout option is secured at the absolute base with proper padding for display safety.Global Auth Reset (5 pts): Activating the logout link completely drops authorization and kicks the user back to the first screen.	
4. Reactive Theme Preferences	20 pts	Dark Mode Switch Logic (5 pts): Tapping the toggle lever smoothly alters the active binary choice via local React states.Instant Theme Overlays (10 pts): Flipping the switch instantly transforms the entire view background canvas into solid black.Contrast Text Integrity (5 pts): System labels automatically convert from black to clear white without font alignment clipping.	
5. Deep Sub-Screen Hierarchy	15 pts	Profile Information Card (5 pts): The profile screen accurately lists your student metadata, university email, and settings card.About Navigation Push (5 pts): Clicking the "About" list button securely pushes navigation forward to reveal the hidden student details.Hidden Page Separation (5 pts): The deep account details file remains accessible via link pushes but is completely invisible on the drawer list.	
FINAL GRADE VALUE	100 pts	Maximum possible score achievable for complete tracking and execution of the specified design flow.	